

SCHEME AND EXCEPTIONS Solutions

COMPUTER SCIENCE MENTORS 61A

November 17 – November 22, 2025

1 Scheme Interpreters

1. Based on the Link class that is below, write the scheme expression represented by the Link class. Assume that this is the Link class in Python that we will be using:

```
class Link:
    '''A Scheme list is a Link in which rest is a Link or nil.'''
    empty = ()
    def __init__(self, first, rest=empty):
        self.first = first
        self.rest = rest

    # There are also __str__, __repr__, and map methods, omitted here.

nil = Link.empty
>>> Link('*', Link(5, Link(Link('-', Link(10, Link(2, nil))), nil)))

(* 5 (- 10 2))

>>> Link('or', Link(Link('>', Link(5, Link(2, nil))), Link(Link('/',
    Link(1, Link(2, nil))), nil)))

(or (> 5 2) (/ 1 2))

>>> Link('+', Link(1, Link(Link('*', Link(Link('-', Link(5, Link(2,
    nil))), Link(Link('+', Link(3, Link(1, nil))), nil))), Link(5,
    nil))))

(+ 1 (* (- 5 2) (+ 3 1)) 5)
```

2. Circle the number of calls to `scheme_eval` and `scheme_apply` for the code below.

```
(+ 1 2)
```

```
scheme_eval    1  3  4  6
scheme_apply    1  2  3  4
```

4 `scheme_eval`, 1 `scheme_apply`.

1. Implement the macro function `combine-num`, which takes in an unquoted list of positive integers and returns a number whose digits are the items of the list in reverse order.

```
; Doctests
scm> (combine-num (1 2))
; 21
scm> (combine-num (2 5 3 5))
; 5352
scm> (combine-num (1))
; 1
scm> (+ (combine-num (1 2 3 4)) 5)
; 4326      # (4321 + 5)
```

```
(define-macro (combine-num lst)
```

```
)
```

```
(define-macro (combine-num lst)
  (if (null? lst) 0
      `(+ ,(car lst) (* 10 (combine-num ,(cdr lst)))))
  )
)
```